SLIME MOULD INSPIRED OPTIMIZATION ALGORITHMS

URBAN PLANNING AND DEEP SPACE EXPLORATION

AARSHIA SHARMA- C440
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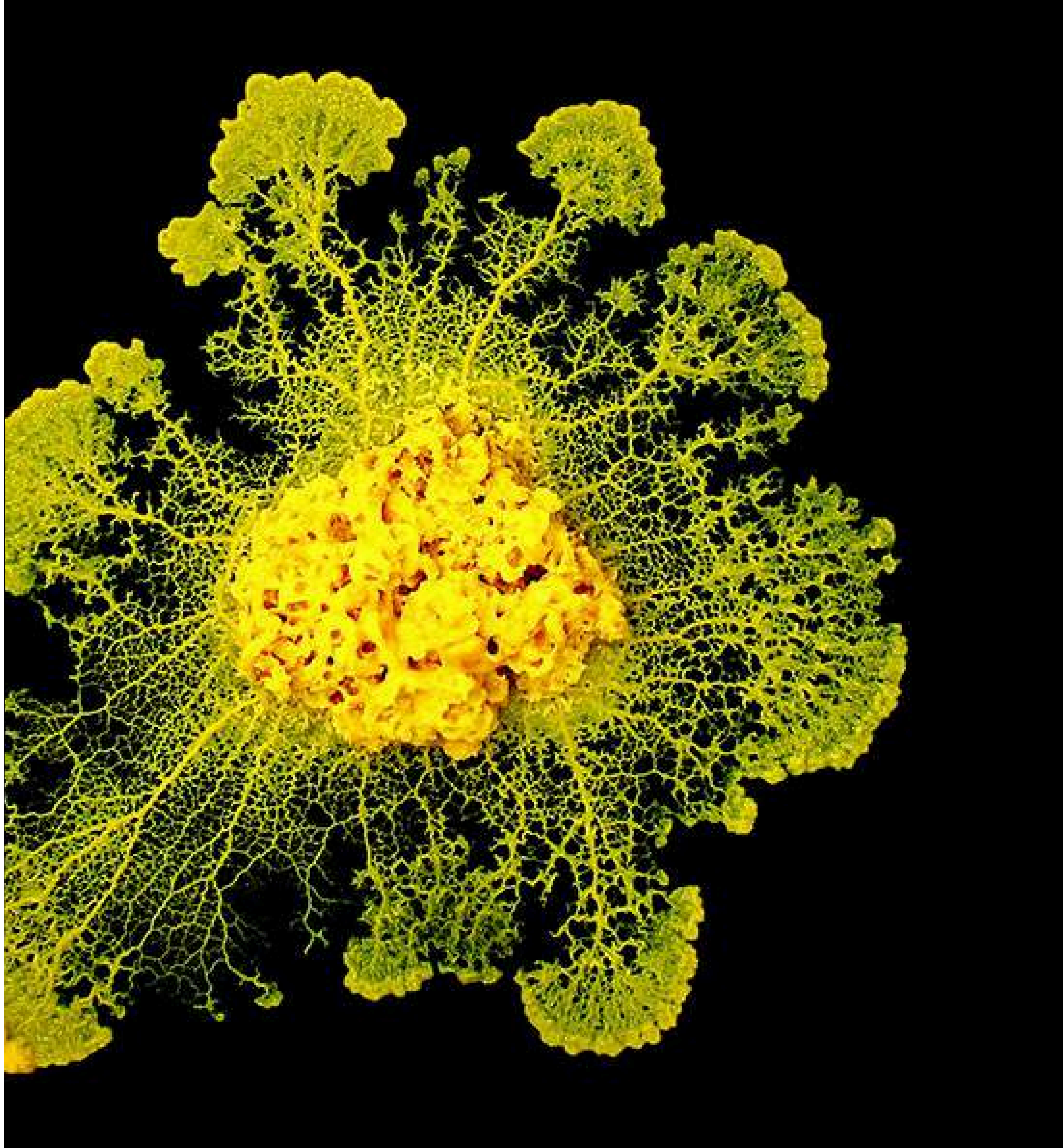

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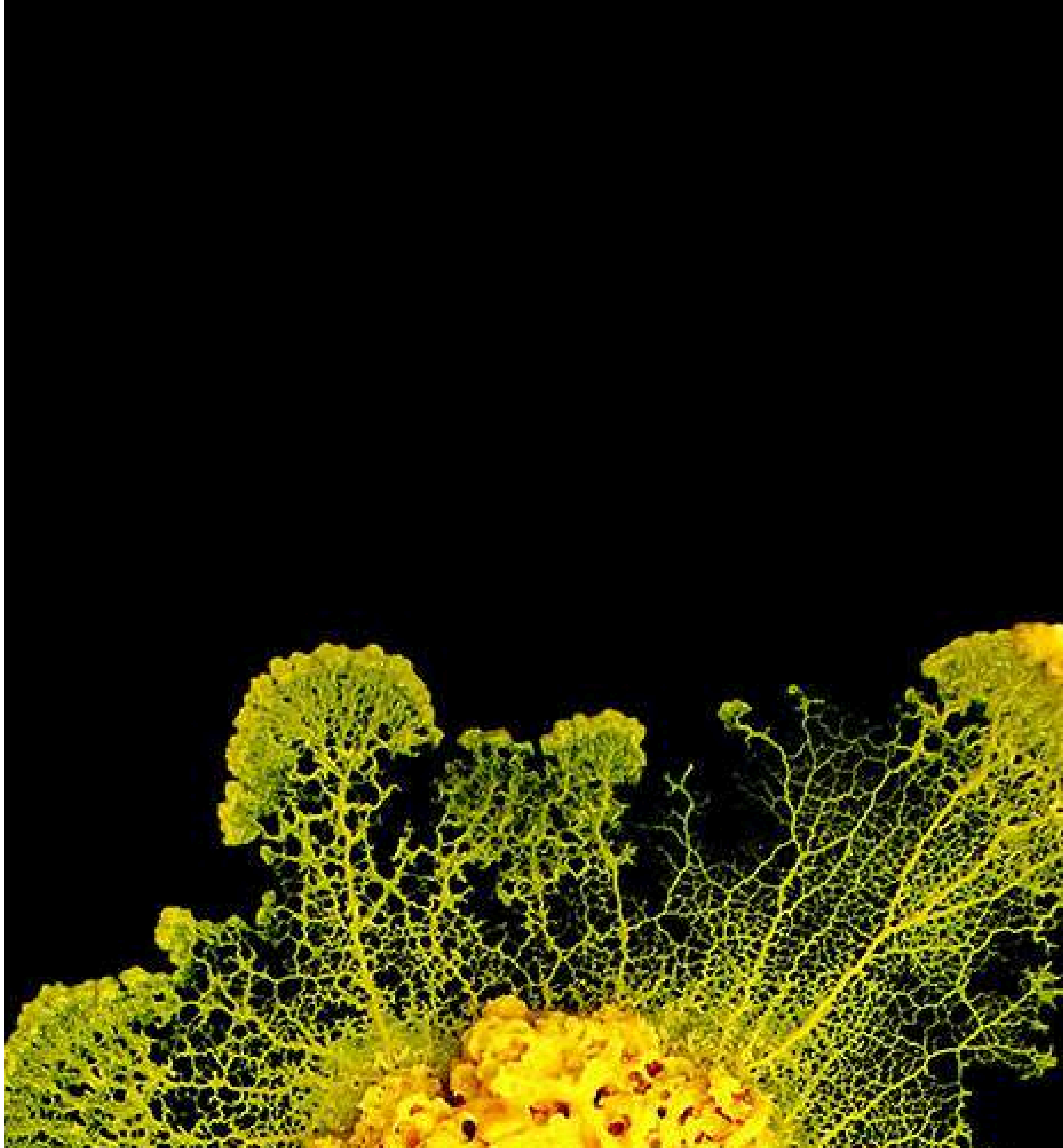
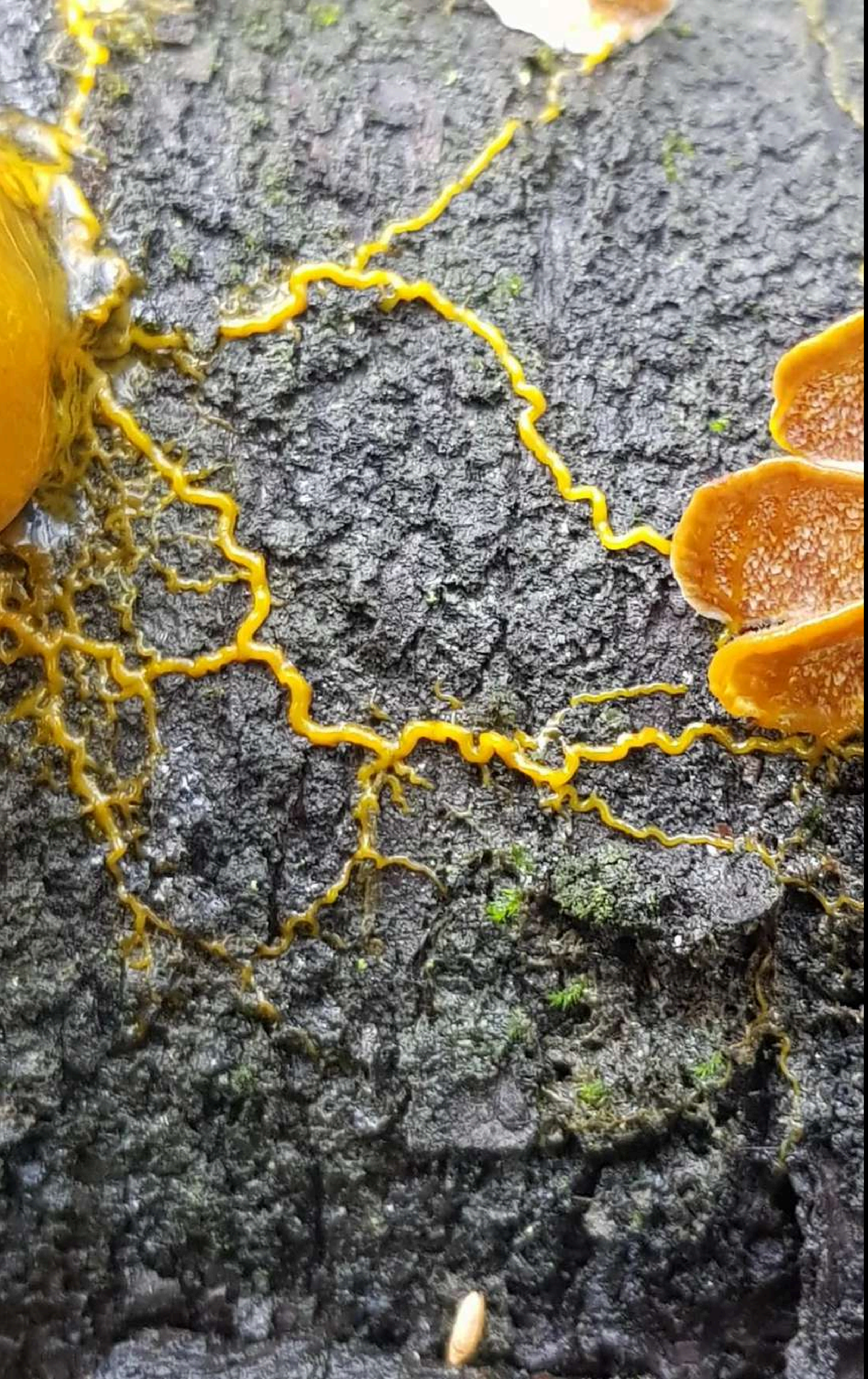
PROBLEM STATEMENT

- Complex network optimization
- Finding shortest and most efficient path
- Balancing cost, efficiency and connectivity





BIOLOGICAL PHENOMENON

Slime moulds are single-celled organisms that can join together toform a large, moving mass called a plasmodium. This plasmodiumcan creep over surfaces in search of food.

TYPES OF SLIME MOLDS

There are two main types:

1. Plasmodial slime moulds: Exist as one giant cell with many nuclei.

2. Cellular slime moulds: Spend most of their lives as separate amoeba-like cells. When food becomes scarce, thousands of cells gather to form a multicellular structure that produces spores.

IMPORTANCE:

- Help decompose organic matter and recycle nutrients.
- Control bacterial populations in soil.
- Widely used to study cell communication, development and cooperative behaviour.

PLASMODIAL SLIME MOULDS

CHARACTERISTICS:
1.Multinucleate
2.No cell walls
3. It moves slowly over surfaces by cytoplasmic streaming, extending finger-likeprojections to search for food.
4. Heterotrophic: It feeds on bacteria, fungal spores, algae, and decaying organicmatter by engulfing them (phagocytosis).
Habitat: Commonly found in moist, shady places such as forest floors, decaying logs,and leaf litter.

LIFE CYCLE:
- Spore formation
- Germination
- Plasmodium stage
- Fruiting body formation

CELLULAR SLIME MOULD

CHARACTERISTICS:
- Unicellular during the feeding stage
- No permanent multicellular body
- Amoeboid movement
- Heterotrophic

LIFE CYCLE:
- Feeding stage
- Aggregation
- Slug stage
- Fruiting body formation
- Spore dispersal

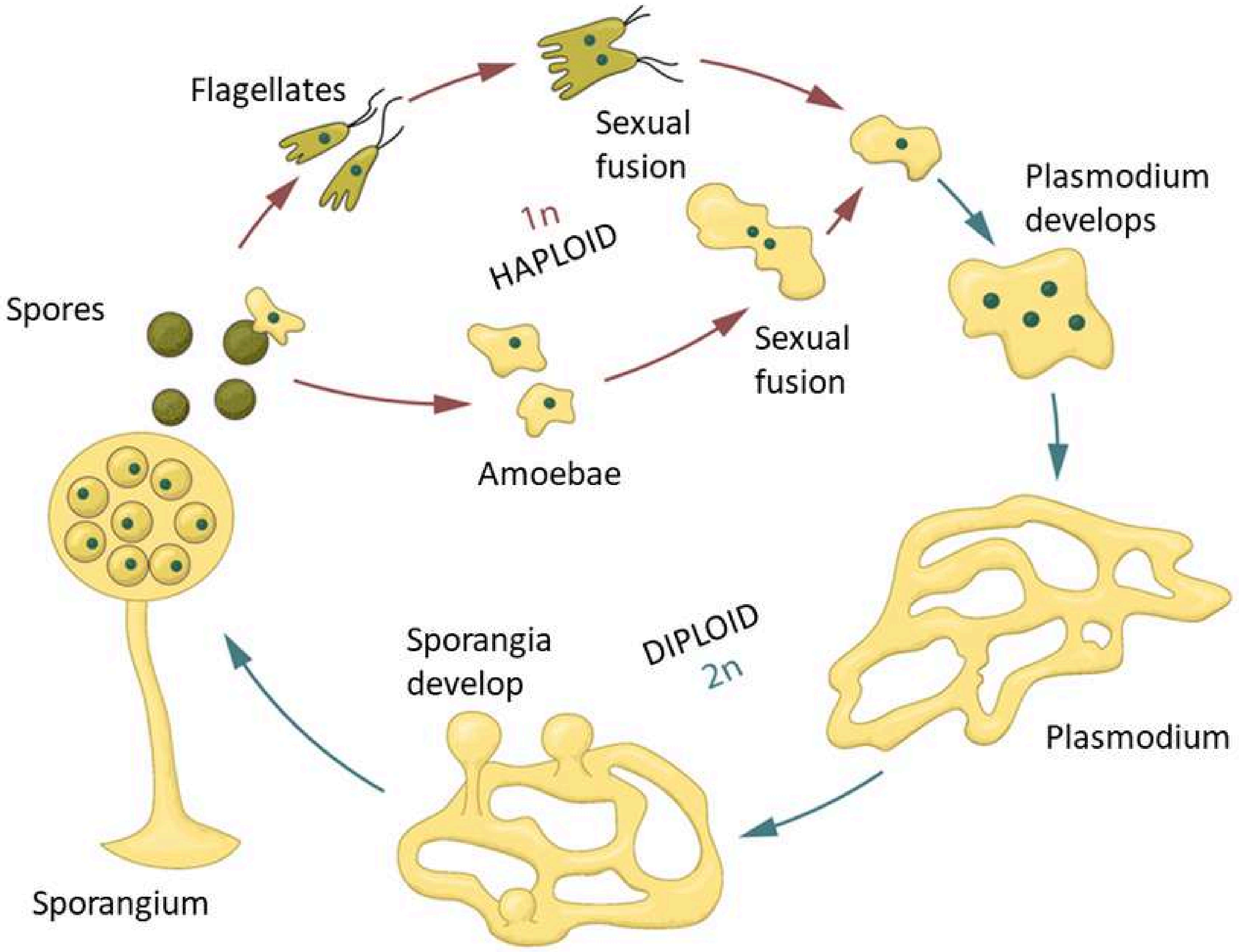
METHODS OF REPRODUCTION

1. ASEXUAL REPRODUCTION

- This is the most common method.
- Under unfavourable conditions, the slime mould forms fruiting bodies (sporangia).
- Inside the sporangia, numerous haploid spores are produced.
- These spores are dispersed by wind, water, or animals.
- When they land in a suitable, moist environment, they germinate into amoeboid cells(or flagellated cells in some species), which grow into new slime moulds.

2. SEXUAL REPRODUCTION

- Sexual reproduction helps create genetic variation.
- Haploid spores germinate into amoeboid or flagellated cells.
- Two compatible haploid cells fuse to form a diploid zygote.
- The zygote undergoes repeated nuclear divisions without cell division, forming a largemultinucleate plasmodium.
- When conditions become unfavourable, the plasmodium develops sporangia.
- Meiosis occurs in the sporangia, producing new haploid spores.



SLIME MOLD LIFE CYCLE

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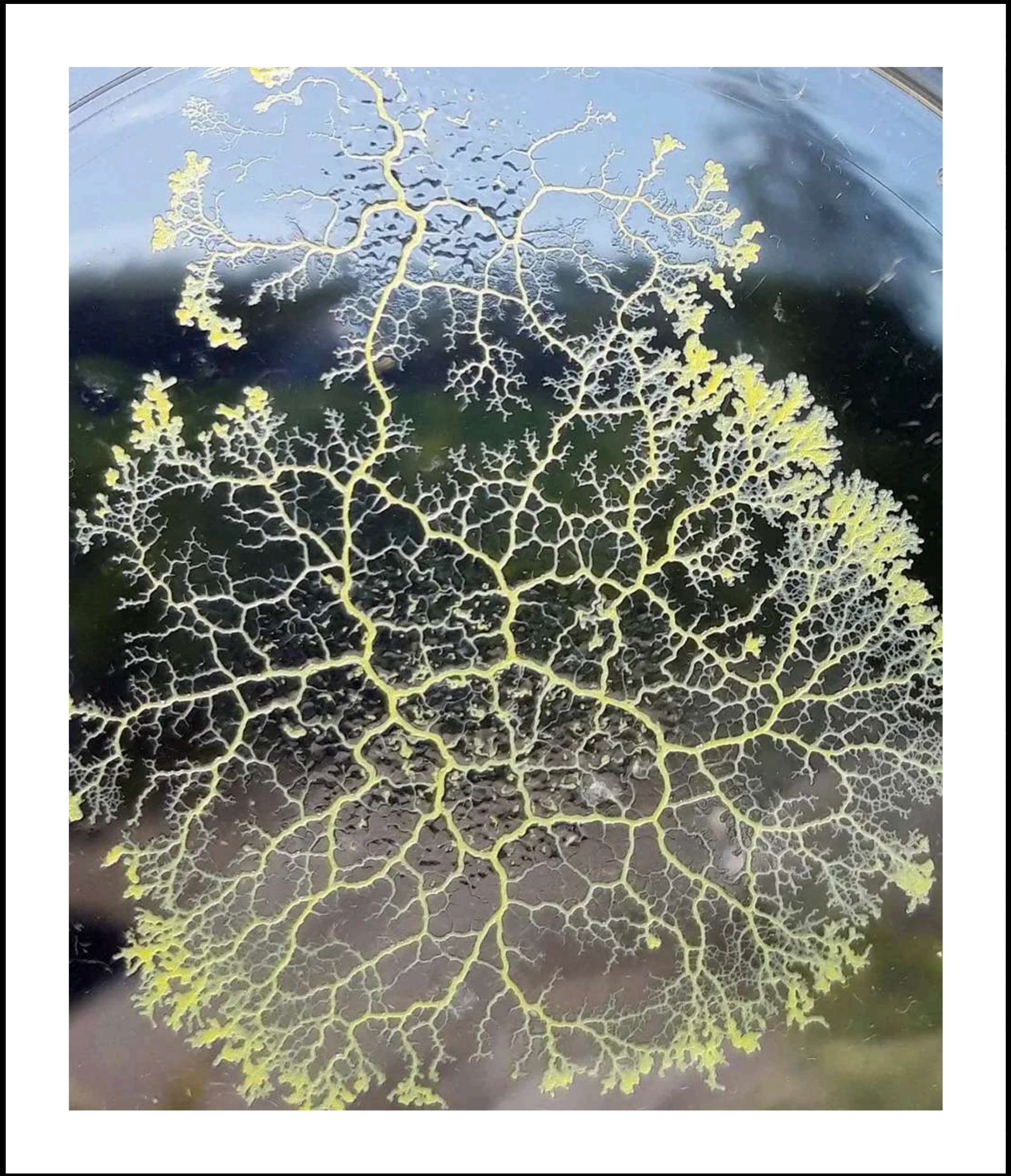
HOW DOES A PLASMODIUM FUNCTION WITHOUT A BRAIN?

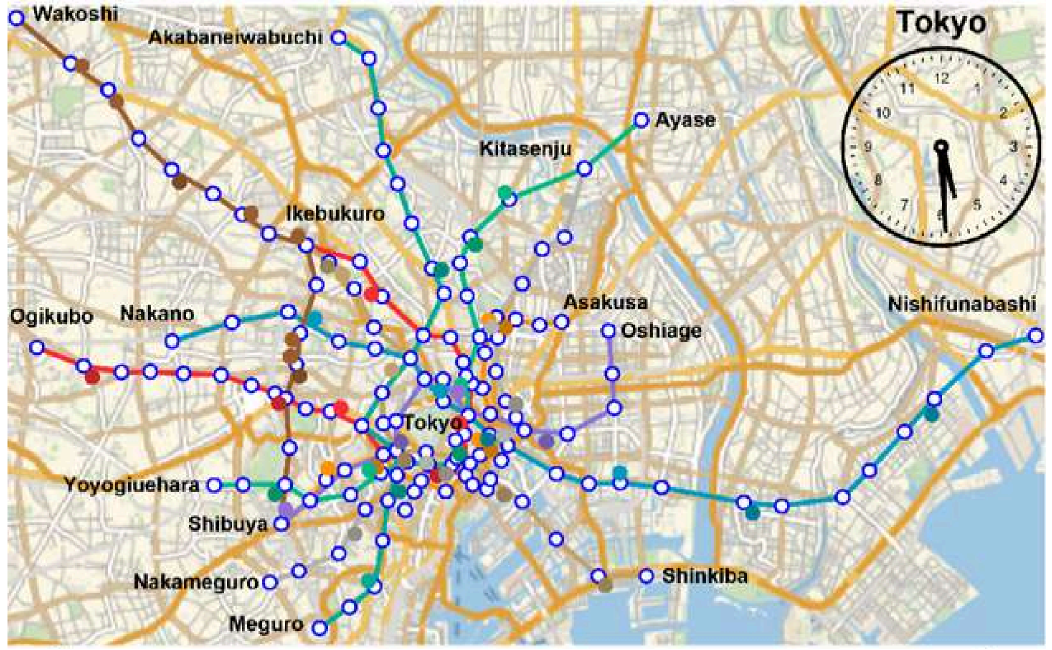
- Biological structure enabling this behaviour
- Single multinucleate cell
- Chemical sensing
- Cytoplasmic streaming
- Dynamic network formation
- When placed in a maze with food at two locations, Physarum polycephalum can exploredifferent paths and eventually forms the shortest, most efficient route. It does not thinkor plan; instead, it uses local chemical signals, feedback mechanisms, and changes incytoplasmic flow to optimize its network.



HOW DOES SLIME MOULD FORM NETWORKS?

1. The plasmodium spreads out
2. Food sources are connected
3. Cytoplasmic streaming strengthens useful paths
4. The network continuously adapts
5. An efficient network emerges

APPLICATIONS- **TOKYO RAILWAY NETWORK**



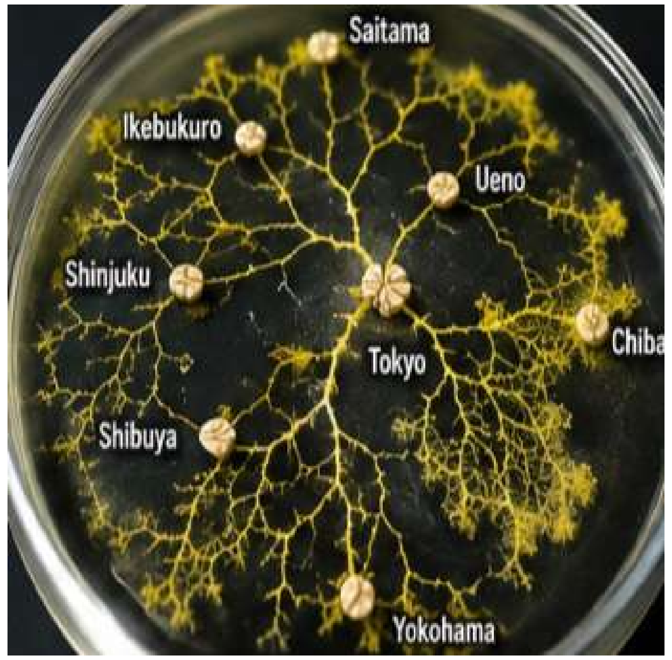
MAP OF TOKYO

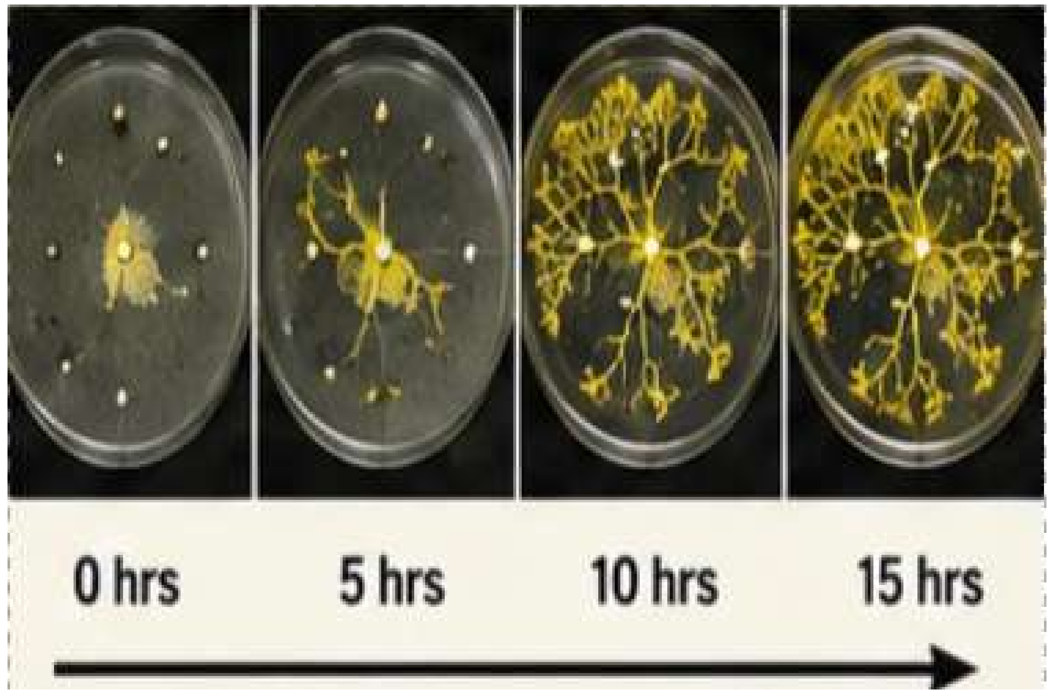
EXPERIMENT

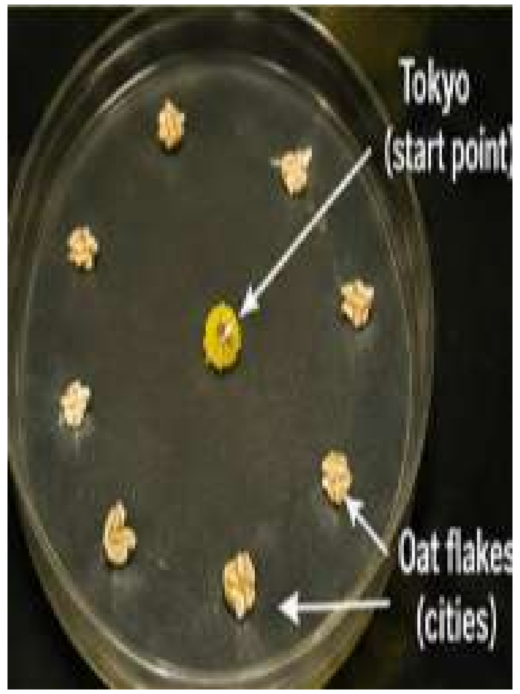
- Oat flakes represented major cities.
- Slime mould was placed at Tokyo.
- It grew naturally towards food sources.
- Over time, only the shortest and strongest paths remained.

RESULT

- Networks closely matched Tokyo's railway system.
- Achieved without any central control.
- Demonstrated efficient route optimization.

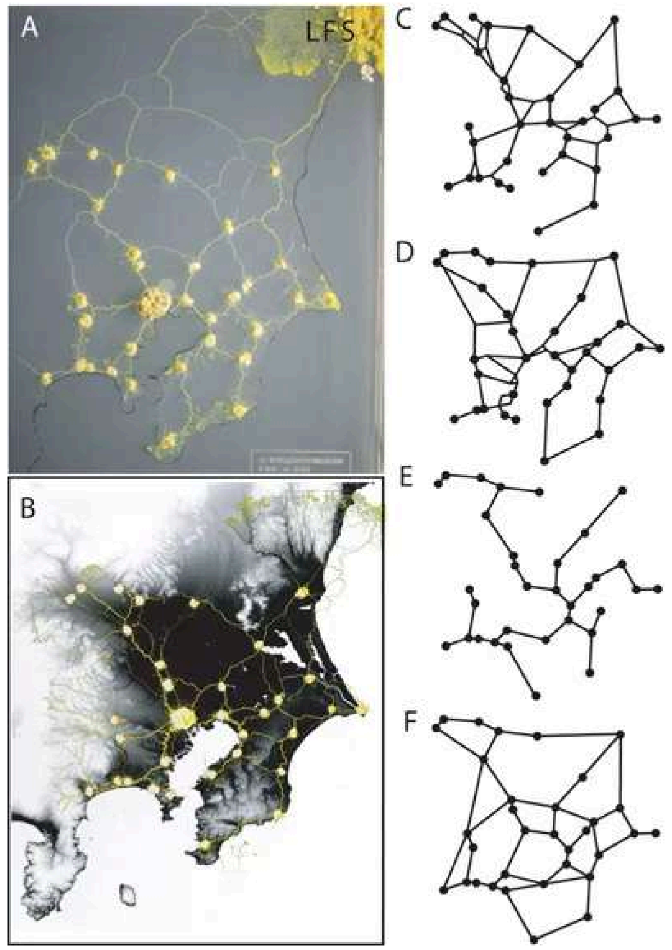






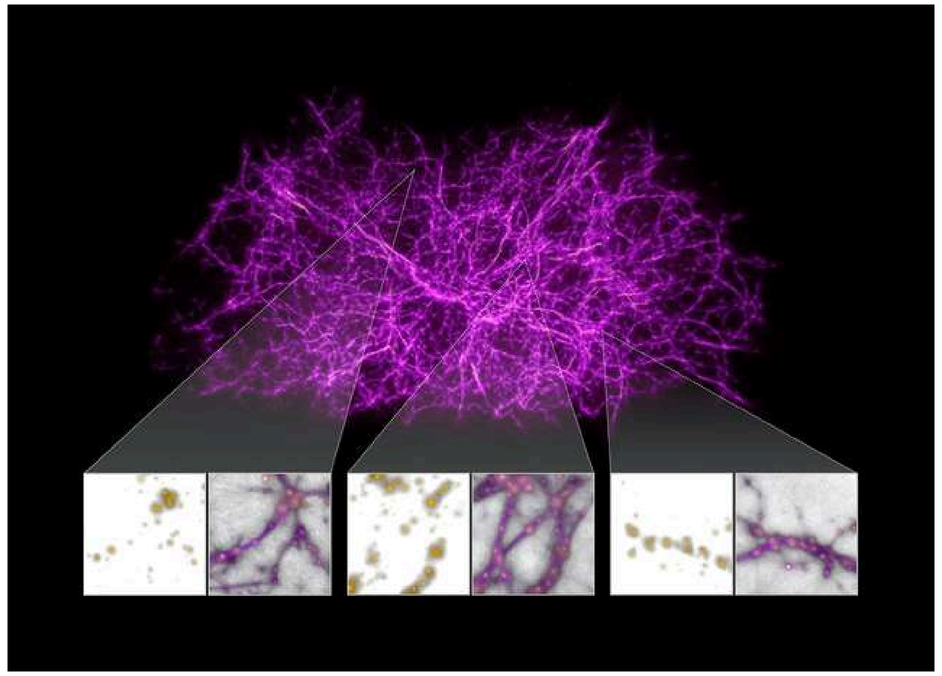
WHY IS THIS USEFUL?

- Shortest routes.
- Cost-effective planning.
- Reliable transportation.
- Inspired smarter transport network design.



Brainless slime mold grows in pattern like Tokyo's subway system | Communicating Science 2017 Section 211

APPLICATIONS- **DEEP SPACE EXPLORATION**



EXPERIMERNT

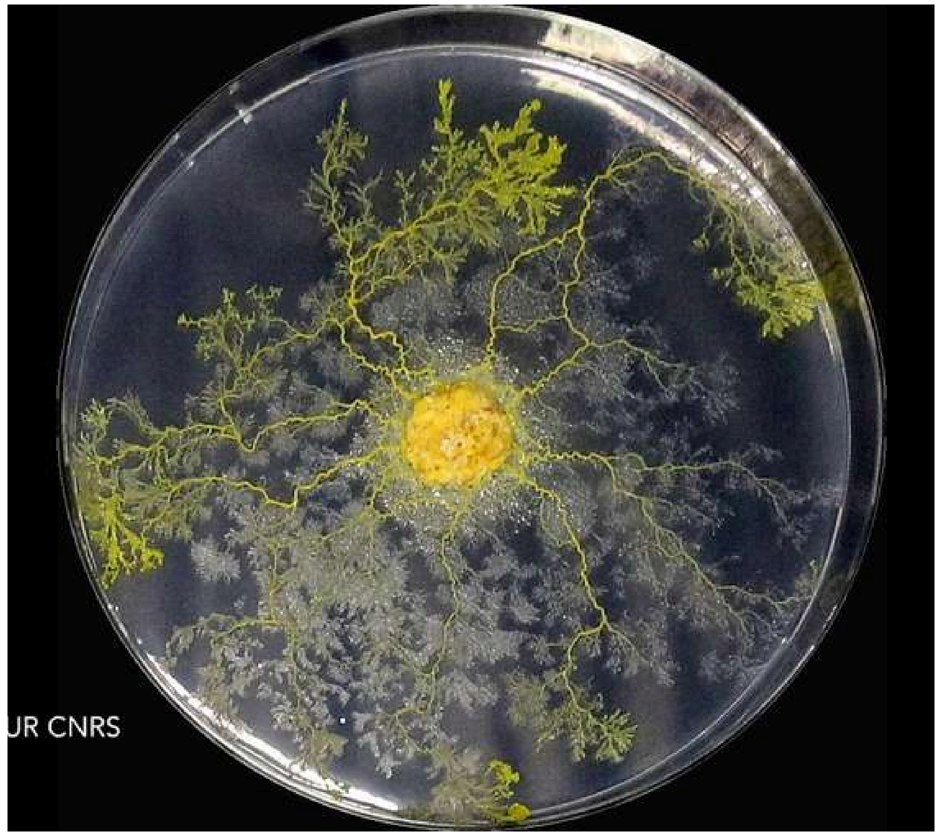
- Researchers developed a slime mould
- Inspired algorithm to
- Model the universe’s cosmic web.
- The alogorithm was tested using dark matter simulations.

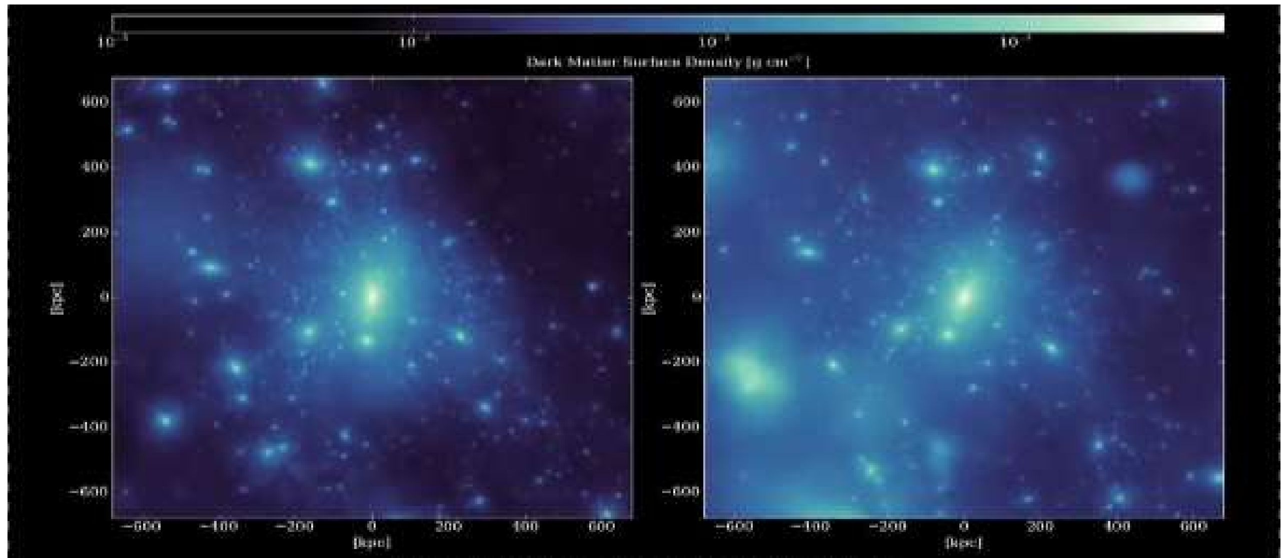
RESULT

- Successfully generated a 3D map of the
- Cosmic web.
- Identified hidden filaments
- Connecting galaxies.
- Produces accurate results in minutes.

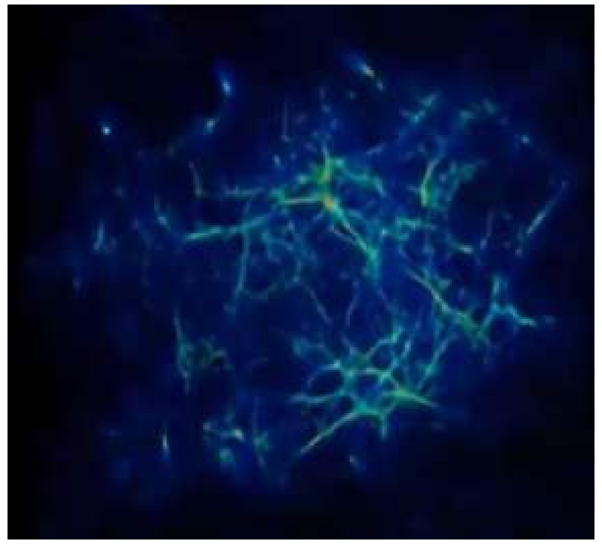
WHY IS THIS USEFUL?

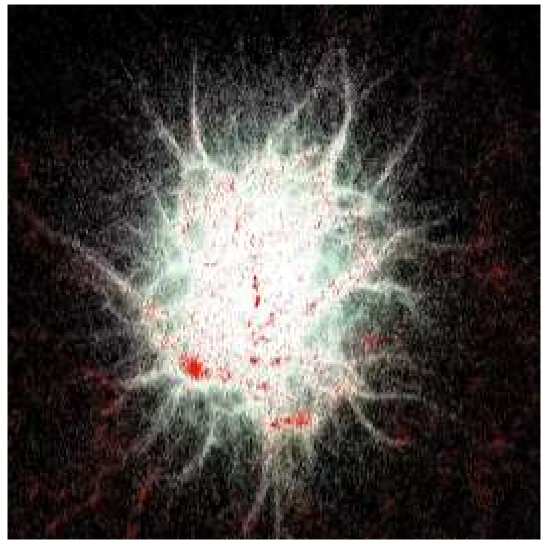
- Maps the large scale structure of the universe.
- Supports future spaceand astronomy research.





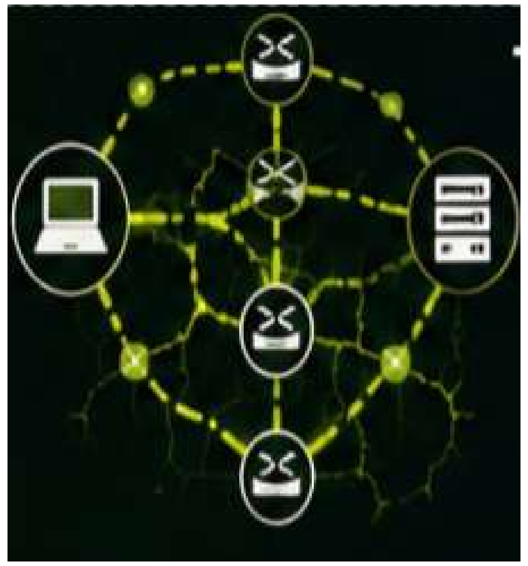
Dark matter density simulation

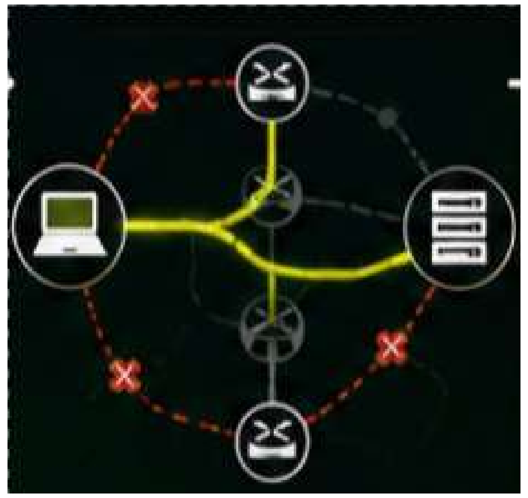




Slime Mold Simulations Used to Map Dark Matter Holding Universe Together - NASA Science

APPLICATIONS- **COMPUTER NETWORK ROUTING**





EXPERIMENT

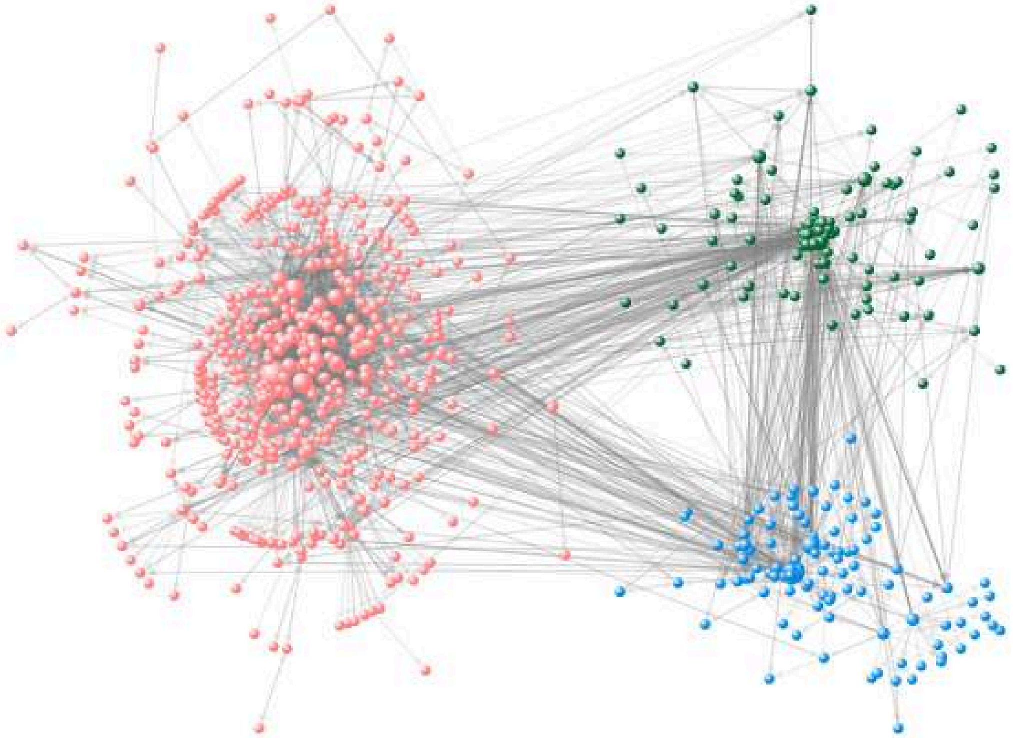
- Simulated a computer network with multiple routers.
- Introduced network congestion and link failures.
- Applied SMA to identify the best communication paths.

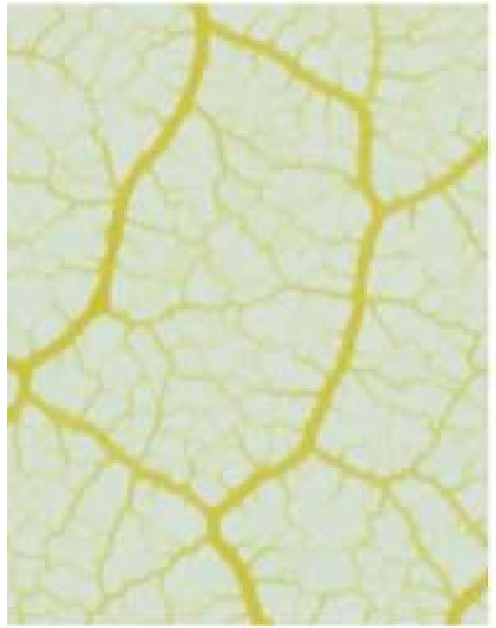
RESULT

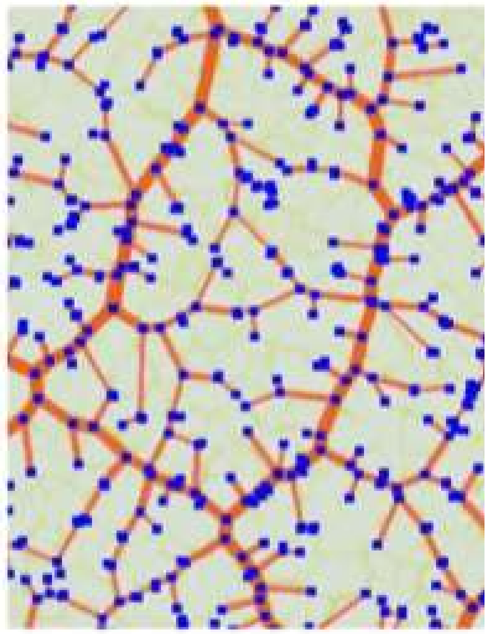
- Selected the shortest and most efficient routes.
- Quickly found alternative paths after link failures.
- Reduced network congestion.
- Improved data transmission speed and routing efficiency.

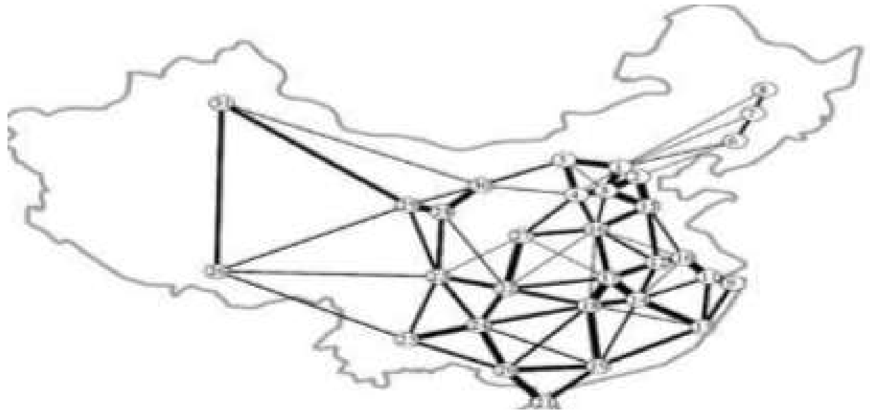
WHY IS IT USEFUL?

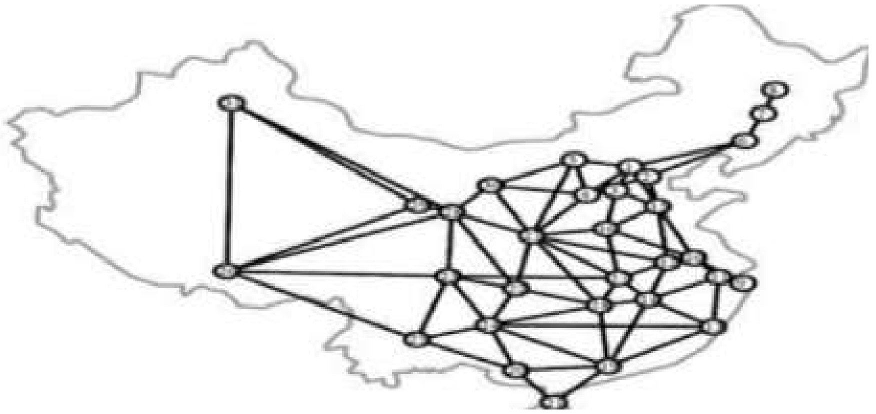
- Ensures reliable data transmission.
- Adapts quickly to network failures.
- Used in cloud computing, data centres and wireless networks.









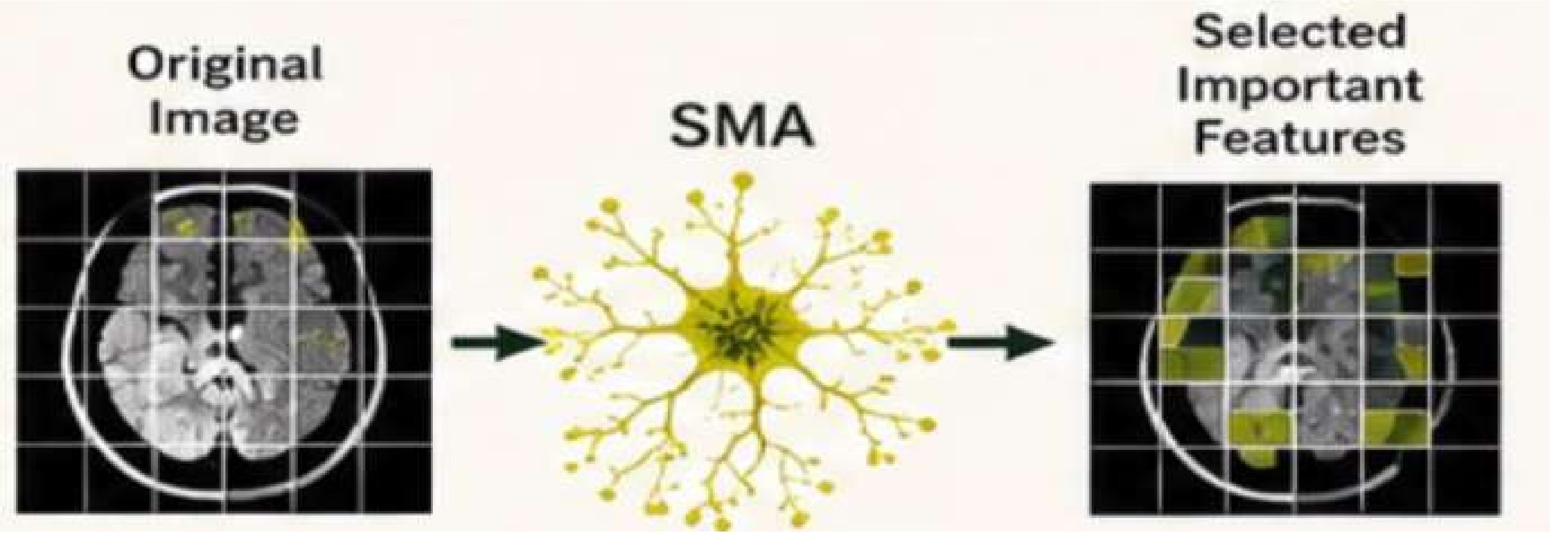


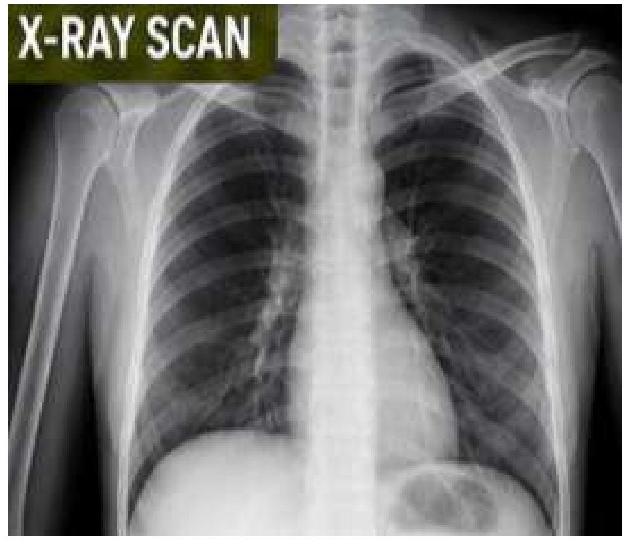
Slime Mold and Network Imaginaries: An Experimental Approach to Communication | Leonardo | MIT Press

APPLICATIONS- **MEDICAL IMAGE ANALYSIS**

EXPERIMENT

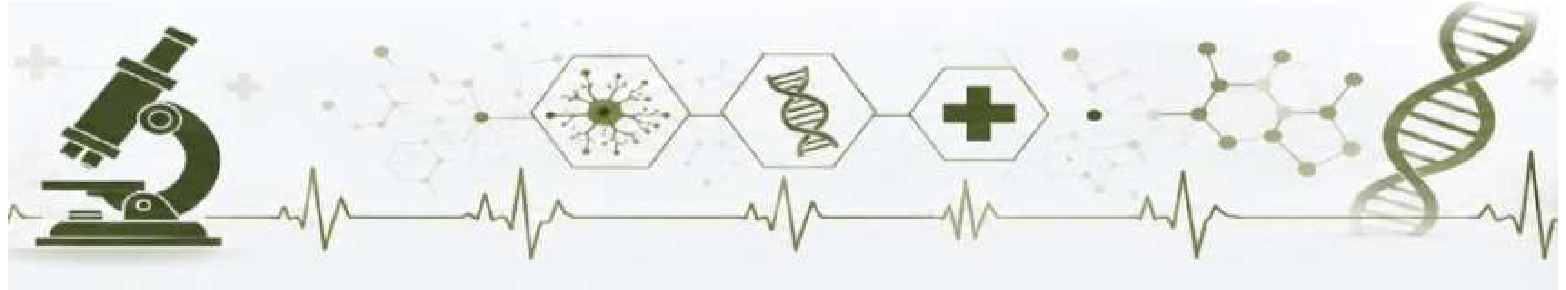
- Tested on MRI, CT and X-Ray datasets
- Slime mold algorithm selected the most important relevant image features.
- Compared with traditional feature selection methods.
- Evaluated for accuracy and processing time

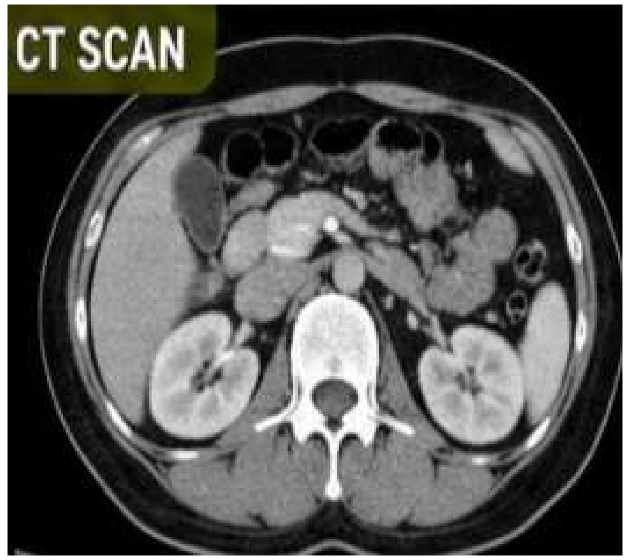




RESULT

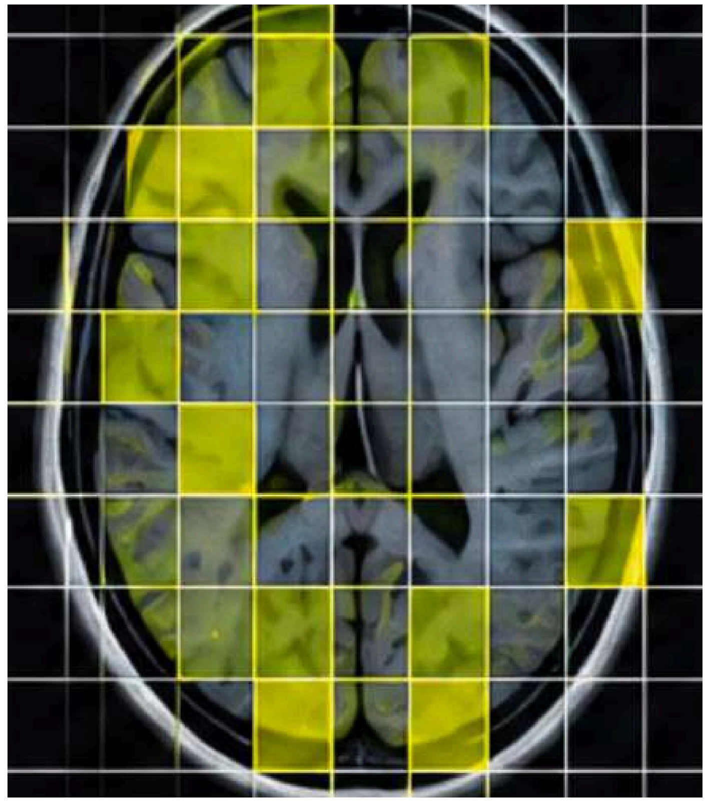
- Faster image processing.
- The algorithm reduces processing time while improving diagnostic accuracy.
- Reduced unnecessary data.
- Enhanced AI diagnosis performance.

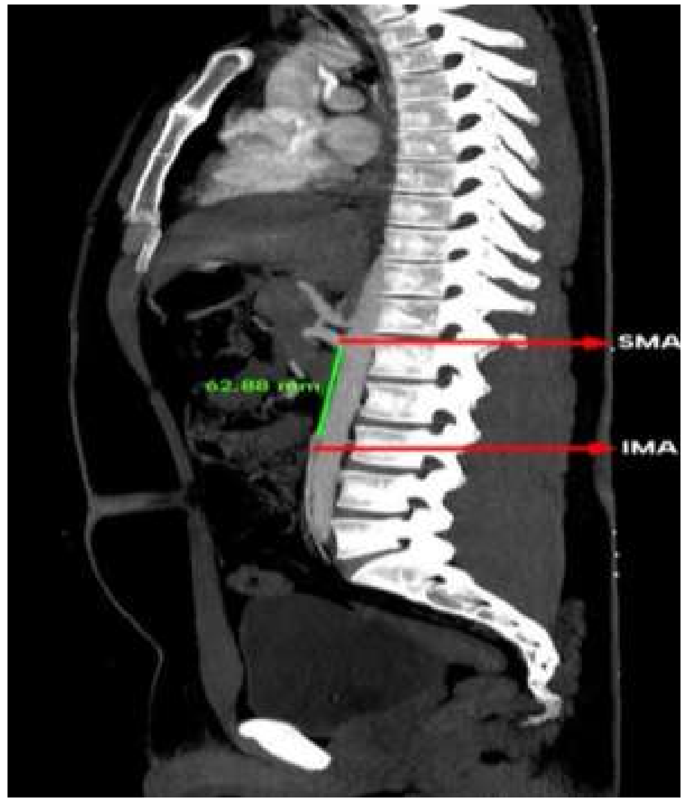




WHY IS THIS USEFUL?

- Faster medical image analysis.
- Higher diagnostic accuracy.
- Reduces computational cost.
- Supports doctors by helping them to focuson the most relevant information.





Slime Mould Algorithm: A Comprehensive Survey of Its Variants and Applications - PMC

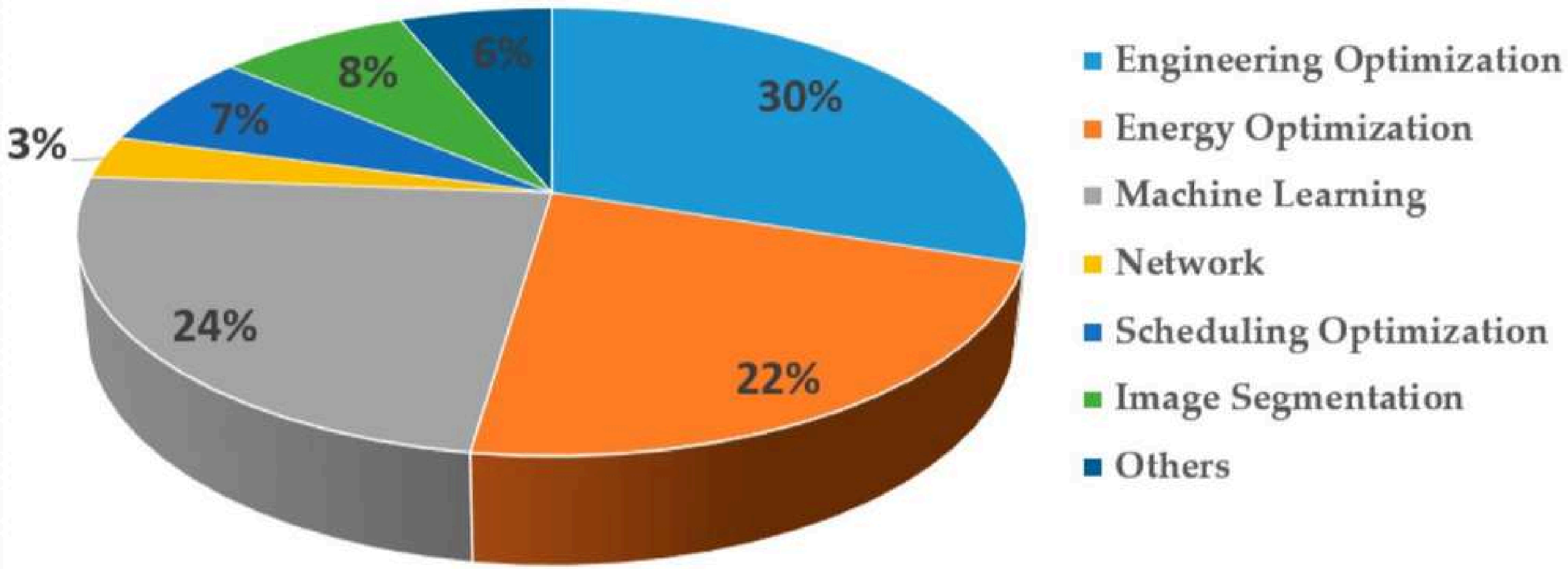
PROS & CONS

PROS	CONS
EXPLORATION AND EXPLOITATION	
Simple structure and easy implementation	Premature convergence
Produces high quality solutions	Performance depends upon the problem
Applicable across multiple domains	Limited performance on high-dimensional problems
Adaptability	Sensitive to population diversity
Supports hybrid improvement	Requires improvement for better performance

STATISTICS

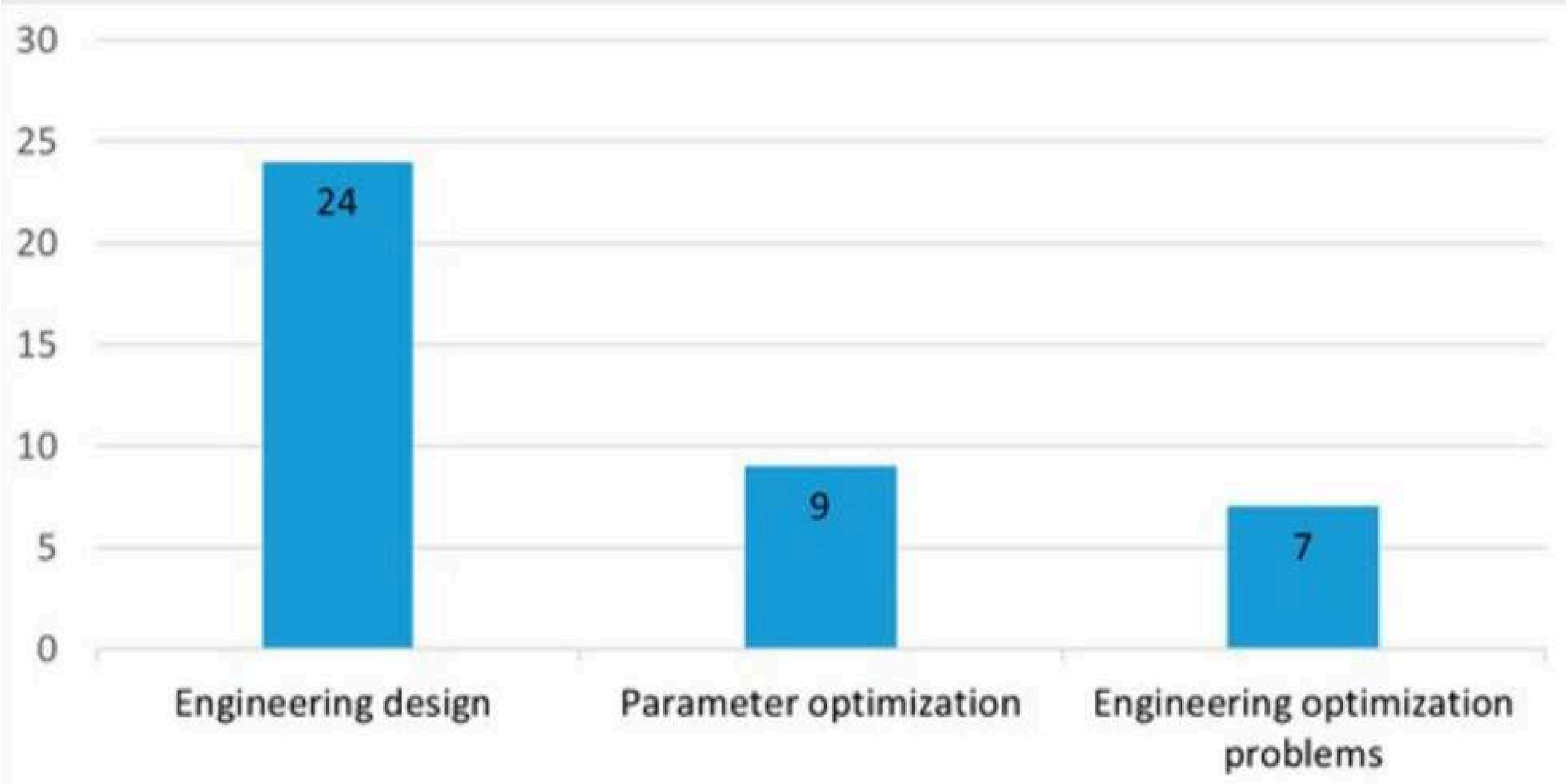
	SLIME MOULD ALGORITHM	ANT COLONY OPTIMIZATION
BASIC PRINCIPLE/ INSPIRATION	Based on oscillation and foraging of slime mould	Inspired by ants searching paths via pheromone trails
CONVERGENCE SPEED	Fast	Moderate
ESCAPE LOCAL MINIMA	Very good	Good
COMPUTATIONAL COMPLEXITY	Medium	Medium-high
PRIMARY ADVANTAGES	Strong exploration and high optimization accuracy	Effective for global search and combination problems
DISADVANTAGES AND LIMITS	Relatively fewer implementation structures documented	High computational cost; poor for continuous optimization

https://www.nature.com/articles/s41598-026-59759-2_reference.pdf



GLOBAL RESEARCH DISTRIBUTION & APPLICATIONMETRICS

Distribution of publications based on their application domains since 2022.



ENGINEERING OPTIMIZATION

Detailed classifications and numbers of the reviewed literature forengineering optimizations.

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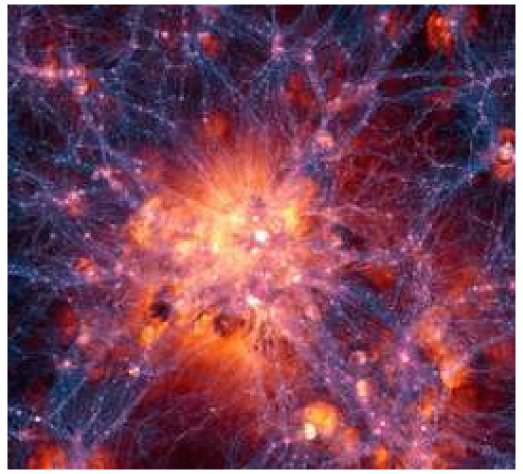
https://www.mdpi.com/2313-7673/9/1/31

RESEARCH TIMELINE

2000
MAZE SOLVING

2010
TOKYO RAILWAY SYSTEM

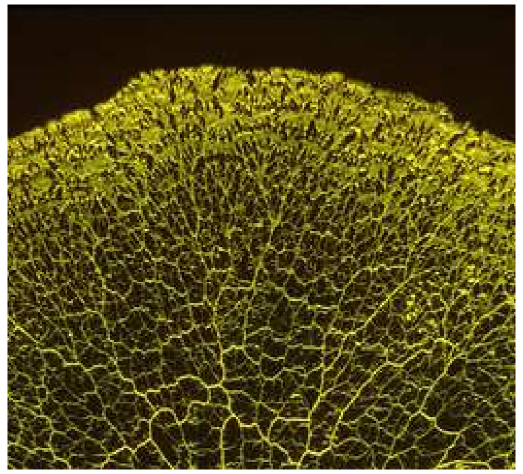
2020
APRIL: Introduction to SMA
MAY: Space exploration



2025-2026
SMART CITY DEVELOPMENT

2008
NETWORK TOPOLOGY

2011-2019
AMEOBA ROUTING METHODS



2021-2024
MULTI-STARTEGY PARAMETER TUNING

PRESENT DAY
AUTONOMOUS INTERSTELLAR TRAJECTORIES

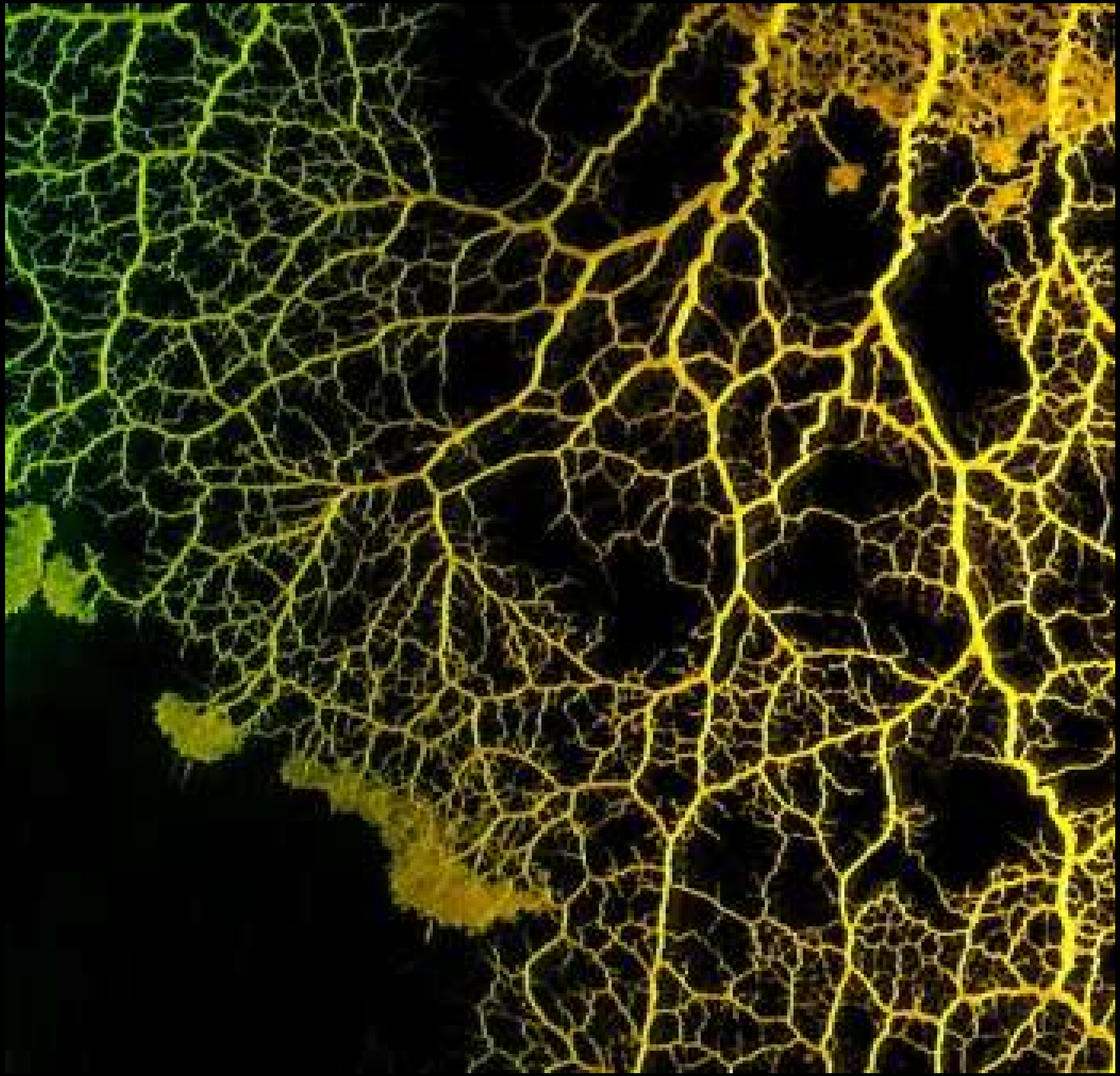
FUTURE POSSIBILITIES

SMART INFRASTRUCTURE AND DESIGN
- urban transit design
- supply chain routing
- telecommunications

ENERGY AND SMART GRID
- power flow optimization
- economic dispatch

ROBOTICS AND AUTONOMOUS SYSTEM
- Dynamic path planning
- Swarm coordination

ARTIFICIAL INTELLIGENCE AND HEALTHCARE
- feature selection
- neural architecture search



SUMMARY

Slime Mould Algorithm (SMA) is a nature-inspired optimization algorithm which simulates the intelligent foraging behavior of slime moulds. This presentation studied the biological phenomenon behind SMA. It discussed how it forms efficient networks without central control and showed its applications in urban planning, deep space exploration, computer network routing and medical image analysis. We also discussed its advantages, its limitations, its research progress and its future prospect. Overall, SMA is a good example of how biological systems can inspire innovative solutions to complex real-world optimization problems and remains a promising approach for future engineering and AI applications.